

Aswinkumar Varathakumaran

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Profile

Undergraduate researcher enthusiastic about computer vision, generative modeling, and multimodal AI, with research experience in real-time perception systems, foundation model adaptation, and applied deep learning for high-impact real world tasks.

Education

Vellore Institute of Technology <i>B.Tech in Computer Science with AI/ML Specialization — CGPA: 8.96</i>	2022 – Present Chennai, India
Delhi Public School <i>CBSE XII — 94.4%</i>	2021 – 2022 Bangalore, India

Experience

Deffo Tech AI AIML Intern	Dec 2025 – Current
Working on India's first fully indigenous Large Language Model developed under the TamilAI project.	
Undergraduate Machine Learning TA 📄	July 2024 – Dec 2024
PI: Dr. Sajidha S A — VIT Chennai	
Assisted final year ML students with machine learning assignments, tutorials, and guided project workflows for hands-on learning.	
Undergraduate Research Assistant	Sep 2024 – Jan 2025
PI: Dr. Nirmala P — VIT Chennai	
Built depth-aware pothole detection system using custom DPT architecture fusing DepthPro and SegFormer for real-time monocular depth estimation and segmentation in ADAS applications.	

Publications

- Varathakumaran, A., & Paramanandham, N.** *Depth-Aware Instance Segmentation for Accurate Pothole Depth Estimation Using Monocular Depth Maps for Advanced Driver Assistance.* **International Journal of Pavement Engineering (IJPE).** (Under Review) 📄 Code 📄 Paper
- Varathakumaran, A., & Paramanandham, N.** *ProvRain: Rain-Adaptive Denoising and Vehicle Detection via MobileNet-UNet and Faster R-CNN.* **ICCID5 2025.** (Presented & In Press) 📄 Code 📄 Paper
- Varathakumaran, A., Jawahara, A., Mynampati, S.A., Sajidha, & Thaseen, S.** *PROXI-NAV: Proximity-Aware Navigation with Audio-Visual Sensing Intelligence.* **FTNCT'08.** (Under Review) 📄 Code 📄 Paper
- Varathakumaran, A., Jawahar, A., & Pandiyaraju, V.** *CLAIMS: Clinical Labeling and Abnormality Inference from Multilead ECG using LLMs with Evidence Citation.* **ICACRS 2025.** (Under Review) 📄 Code 📄 Paper

Projects

Intent-Aware Clinical Product Recommendation System (ClinicalBERT, Sentence Transformers, FAISS, NoSQL, PubChem API, RxNorm API)	Jan 2026
- Built an NLP-driven clinical product recommendation system at the Apollo THIT Solvathon 2026 by extracting therapeutic intent and contraindications from doctor patient conversations into structured JSON. - Implemented semantic retrieval over the Apollo Catalog using Sentence Transformers and FAISS with NoSQL-based filtering to exclude contraindicated ingredients and recommend exact and class-equivalent drug alternatives. - Normalized all drug and ingredient entities using PubChem and RxNorm APIs for standardized clinical reasoning.	
Space Agriculture Data Portal 🌐	Feb 2025
(NASA OSDR API, LangChain, Ollama, FAISS, NoSQL, FastAPI, Streamlit)	
- Built a data exploration portal for NASA OSDR, automating dataset ingestion, metadata parsing, and time-series analysis of microgravity plant-growth experiments. - Implemented semantic retrieval and question-answering using LangChain, Ollama, and FAISS over experiment metadata and mission logs. - Developed a Streamlit dashboard backed by FastAPI for interactive visualization of growth curves, environmental parameters, and genotype comparisons.	
IMU-Guided Obstacle Prediction for Robust ADAS (Sensor Fusion, ADAS, IMU Sensors)	Jan 2026
Designed a camera-IMU fusion framework to predict obstacle locations during vision degradation caused by high-dynamic vehicle motion.	

- Used IMU-based ego-motion estimation to propagate obstacle states from the last reliable camera frame and reproject predictions into image space with uncertainty modeling.
- Validated on autonomous driving datasets under simulated camera degradation, demonstrating improved obstacle retention compared to vision-only baselines.

Basketball Player Tracking with Court Projection

May 2025

(YOLO, Streamlit, Docker, FastAPI)

- Developed a complete sports analytics pipeline using YOLOv11 trained on custom DeepSportRadar court and player annotations, enabling robust multi-player tracking under dynamic lighting and motion.
- Computed homography transformations to map broadcast-view detections onto a top-down court representation, enabling zone-based heatmaps, possession analytics, and trajectory inference.
- Built a Dockerized Streamlit-FastAPI deployment supporting live video ingestion, asynchronous inference, and real-time overlay rendering with sub-80ms latency per frame.

Achievements

- Program Representative for **Computer Science AI&ML branch** at *VIT Chennai*.
- Top **5 Finalist** at *THIT Apollo Solvathon 2026* a 36-hour hackathon with **140+ participants** nationally.
- Top **10 Finalist** at *Hack The Horizon 2.0* — VIT Chennai AI Club's flagship 24-hour hackathon with **200+ participants** from South India.
- Top **5 Finalist** in *Cosmonova*, a space-based research hackathon hosted by **SEDS Antariksh**.

Extracurriculars

- **Treasurer & Marketing Lead, Socrates Club - VIT** Aug 2023 – Present
Managed club finances, sponsorship acquisition, budgeting for events, directed promotional strategies, designed materials, and enhanced campus visibility.
- **Sophrosyne VIT, Toastmasters International** July 2024
Was part of the newsletter editorial team and organized public speaking workshops for 80+ participants.
- **Internshala Student Program (ISP) 2025** Mar 2025
Represented Internshala, promoting internships, trainings, and student engagement.
- **Marina Beach Cleanup** May 2024
Co-led 3 km cleanup with 15+ volunteers.
- **American Football Team — VIT Chennai** June 2023 – April 2024
Vice Captain, led a 15-member team to semi finals in a state level tournament.

Certifications

Advanced Computer Vision with Tensorflow — <i>Coursera, DeepLearning.AI</i>	Sep 2025
Oracle Cloud Infrastructure 2025 Generative AI Professional — <i>Oracle University</i>	June 2025
Neural Networks and Deep Learning — <i>Coursera, DeepLearning.AI</i>	Aug 2024
Forge Industry Certification – Accenture Nordics — <i>Forage</i>	May 2024

Technical Skills

- **Languages:** Python, C/C++, Java, SQL, HTML/CSS, JavaScript
- **Frameworks:** PyTorch, TensorFlow, Streamlit, React, Next.js
- **Developer Tools:** Git, VSCode, CVAT, Roboflow, Docker
- **Libraries:** OpenCV, MediaPipe, HF Transformers, LangChain, pandas, NumPy
- **Vision Models:** YOLO, Detectron2, SAM, HRNet, MiDaS, ZoeDepth
- **Core:** Keypoints, Segmentation, Homography, Depth Estimation, Video Inference
- **NLP:** NER, Sentiment Analysis, Speech-to-Text, Text Generation
- **APIs & Data:** HF API, NASA OSDR API, REST APIs, SQLite3, MySQL